

PRADEEP JAJARA

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Summary

Lead Forward Deployed AI Engineer with 5 years of experience building production-grade multi-agent systems using Google ADK and Gemini, with 6+ custom MCP servers deployed in Azure AKS serving 500+ daily requests at 99.9% uptime. Built 4 autonomous multi-agent systems with LLM-as-Judge evaluation, critic-refiner patterns, and HITL workflows for compliance. Specialized in agentic AI orchestration, enterprise MCP infrastructure, and multimodal document processing with proven 90% time reductions across production environments.

EDUCATION

2016 - 2020 | B.Tech. in Computer Engineering

Indian Institute of Information Technology, Design, and Manufacturing,
Kancheepuram, Chennai, Tamil Nadu, India

PROFESSIONAL EXPERIENCE

Neurealm (formerly GAVS Technologies), Chennai | May 2019 – Present

Lead AI Engineer | July 2024 – Present

Immigration Automation Platform (Forward Deployed Engineer — UAT)

- Built end-to-end AI-powered immigration document management system for enterprise HR team tracking 130–160 active H1B/L1 visa cases—replacing 7–10 hours/week of manual processing with AI-assisted verification (currently in UAT, pending leadership review).
- Implemented LLM-as-Judge (Gemini via Vertex AI) to verify scanned immigration documents (I-797, L-1, Passport)—AI proposes document verification verdict with written justification, human approves via HITL workflow ensuring compliance integrity.
- Built amendment portal with secure magic-link employee document uploads, smart document reuse (only requests missing docs per case), and automated email notifications—eliminating 3–4 rounds of manual follow-up per amendment case.
- Developed compliance risk dashboard with pivot-style status filtering across 100+ employee records—surfacing hidden risk cases (extension pending, amendment pending, GC status) replacing manual Excel-based tracking.

AI-Powered Work Item Generation System

- Architected production-grade multi-agent system using Google ADK and Gemini with 4 autonomous agents (DocumentProcessing, WorkItemGeneration, Evaluator, Critic)—reduced manual work item creation from hours to minutes (90% time reduction).
- Implemented LLM-as-Judge evaluation framework with 4 research-backed rubrics (Coverage 35%, Completeness 30%, Traceability 25%, Hallucination Detection 10%), validated against IEEE/ISO standards.

- Engineered a parallel document processing pipeline using dynamic agent spawning and async/await patterns—processes documents concurrently, eliminating sequential bottlenecks.
- Implemented HITL (Human-in-the-Loop) workflow via Athena for manual review before Azure DevOps integration—ensures complete traceability for regulatory compliance (FDA, ISO 26262, DO-178C).
- Built CriticAgent for automated requirements gap analysis with iterative improvement workflow (Analyze → Update → Re-evaluate).

Model Context Protocol (MCP) Server Infrastructure

- Architected and deployed 6 custom MCP servers in Azure AKS clusters with API Management for enterprise-grade access control—Azure DevOps MCP, Athena Custom MCP, Git MCP, and domain-specific tools.
- Engineered MCP infrastructure using Docker, Kubernetes, and Azure API Management—achieved high uptime serving 500+ daily agent requests.
- Enabled agents to dynamically discover and invoke 50+ tools through a unified interface, eliminating custom integrations per tool.
- Built an Azure DevOps MCP server supporting 15+ operations integrated with HITL workflow for mission-critical project management.

SDD Generation System

- Architected end-to-end multi-agent system with 4 specialized agents using Sequential, Parallel, Loop patterns—reduced review iterations by 40%, manual effort from days to minutes.
- Implemented critic-refiner pattern using LoopAgent—eliminated 70% of manual revisions through automated refinement loops.
- Integrated multi-cloud orchestration (AWS S3, GCS, Vertex AI) with Confluence API automation—processes 2GB+ documents autonomously at high uptime.

Software Engineer | April 2021 – June 2024

Automation using Python and Apache Airflow

- Developed automation solutions using Python and Apache Airflow—built self-healing workflows with automatic retry mechanisms, managing 500+ daily tasks across 10+ production workflows.
- Integrated Kubernetes, MongoDB, and PostgreSQL for robust data handling—built monitoring dashboards using Prometheus and Grafana, reducing MTTR by 45%.

Data Science Intern | May 2019 – October 2019

- Built a contextual chatbot using ML and NLP (NLTK, Spacy, Gensim) deployed via Flask API—served 1000+ user queries with 85% intent classification accuracy across 50+ intent categories.

PROJECTS

NSFW Text Classification

- Developed a machine learning model for classifying Reddit post titles as NSFW (Not Safe for Work), enhancing content moderation processes using NLP techniques—deployed a binary classification system with 92% accuracy across 100K+ samples.
- Utilized Python, Pandas, Scikit-Learn, NLTK, Gensim's Word2Vec, and Hugging Face Transformers to preprocess data, generate embeddings, and perform classification—implemented an ensemble approach combining traditional ML and transformer models.

Suicide Tweet Identification - Hackmakers Hackathon

- Collected data on tweets related to suicide before and after COVID-19 using Twitter APIs and web scraping—built a dataset of 50K+ tweets with temporal analysis and sentiment labeling.
- Analyzed data and created AI models to classify tweets according to risk level with 88% precision—implemented multi-class classification (Low Risk, Medium Risk, High Risk, Crisis) using BERT-based transformers.
- Secured 3rd place in hackathon for innovative approach to mental health crisis detection—developed early warning system for social media monitoring with explainable AI insights and real-time alerting capabilities.

Aqua Culture Monitoring using ML, DST-sponsored, Govt. of India

- Developed an efficient water quality monitoring system by minimizing the number of sensors through predictive modeling—reduced sensor requirements by 40% while maintaining 93% accuracy in water quality predictions.
- Predicted relations between attributes and identified faulty readings using ML models (Random Forest, XGBoost)—implemented anomaly detection for sensor malfunction identification with an automated alerting system.

SKILLS

- AI & Multi-Agent Systems: Google ADK, Multi-Agent Orchestration, Model Context Protocol (MCP), LLM-as-Judge, RAG
- Programming: Python, NodeJS
- Cloud Platforms: Azure, GCP, AWS, Docker, Kubernetes
- ML/DL Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, CrewAI
- Data Engineering: Vector Databases(Qdrant, FAISS), Apache Airflow, MongoDB, PostgreSQL

TECHNICAL CERTIFICATIONS

- Natural Language Specialization (Deep Learning.ai & Coursera)
- Data Analysis with Python (IBM & Coursera)
- Deep Learning Specialization (Deep Learning.ai & Coursera)
- Python for Data Science and Machine Learning Bootcamp (Udemy)

ACTIVITIES AND AWARDS

- NeuStar - Star Performer for Quarter 2026 Q1 at Neurealm.
- Star Performer for Quarter 2022 Q1, Quarter 2024 Q1 at Neurealm(formerly GAVS Technologies).
- Star Team of the Year 2021-2022, 2023-2024 at Neurealm(formerly GAVS Technologies).
- Secured 3rd Place in Hackmakers Build With AI Hackathon
- Secured 1st Place in AI Agent Marathon at Neurealm for building a code review agent
- Secured 1st Place in AVISHKAR Ideathon at IIITDM Kancheepuram